

Theory and procedures of manufacturing a cast nozzle

Translated Technical Report

Theory of manufacturing methods

There are multiple types of manufacturing methods:

1. **turning** (by removing material on a lathe, cnc or similar device the final product is gradually formed)
2. **casting** (a homogeneous mass is poured into a negative /mold/ and after hardening the final product is removed)
3. **layering** (layers of materials are applied to a negative, which subsequently form the final product)

We decided to choose casting, as it is a simple and cheap method of manufacturing durable parts.

Tested and used materials

Ideal material

By the term ideal material we understand a material that possesses ideal properties for the given application. In our case, the ideal material is a material with low weight and high thermal and pressure resistance. Also it is easily processed (cast).

Consideration

There are many materials on the market that can be cast. An ideal choice would be a material like refractory concrete, which possesses high thermal resistance up to 1700°C. This material is however expensive, sold only in practical 25 kg packages and dense. Therefore we considered the use of more accessible materials, specifically a mixture of sand, plaster and concrete mixture (hereinafter referred to only as concrete).

Our consideration:

1. **concrete mixture** - will provide strength and heat resistance
2. **plaster** - will lighten the mixture and fill small and complex geometry of the mold
3. **sand** - will provide strength, as it acts as a medium for the concrete

Tested mixtures

We subjected to a test multiple ratios of the already mentioned three components (hereinafter also as three-component). We tried to achieve a ratio whose final measured quantities will most closely meet the definition of an ideal material. Among the tested mixtures belong:

Sample	Ratio [m]	Material
v1	13:26:7:40	plaster:knauf_concrete:water:sand:water
v2	12:5	knauf_concrete:water
v3	35:35:22	knauf_concrete:sand:water
v4	24:5	fofr_concrete:water

By previous tests we came to the conclusion that V4 is the ideal mixture. The use of plaster, as mentioned in the consideration, did not occur, as it had adverse effects on the final strength of the material.

Used accessories

Theory and mold design

When designing a mold for casting a quality-compliant part, it is a necessity to keep in mind the following:

1. air exhaust
2. simplicity of pouring the mixture
3. simplicity of pulling out the casting
4. surface finish of the mold and casting

Guided by these rules we design a mold:

- with holes through which air escapes
- with holes large enough for pouring the raw mixture
- with the possibility of its easy disassembly

Our mold consists of three parts, 1. converging (red), 2. diverging + throat (blue) and 3. outer shell (green) as can be seen in the picture. Parts 1 and 2 we printed on a 3D printer. For 3 we recommend using a non-stick material and/or one must count on the degradation of the part.

Mold fabrication

We printed the mold on an FDM 3D printer, while we chose a vertical orientation of the print to simplify the printing process. The use of other types of 3D printers, such as SLA, is certainly suitable, perhaps even more reasonable.

Mixture vibrating (device)

We built the vibrating device from four springs, wooden boards, an AC voltage regulator and a universal carbon motor. Furthermore, suitable connecting material was used.

Release solution

It is a solution that will allow easier removal of the casting from the mold. In our case we used a mixture of water, IPA and solid soap. We saturate hot water with solid soap and pour in an equal part of IPA. We apply the mixture in 2-4 layers with a brush and let it dry.

Theory and procedure of casting

Theory

During casting we must pay attention to the correct time sequence of the performed actions. Furthermore we must pay attention to the time after which the mass will be too stiff for casting.

Procedure

We wrap the outer shell of the mold from the inside with one layer of office paper and from both sides we catch it with transparent adhesive tape. We coat the converging and diverging part with the release layer. We connect the parts with an M6x60 screw. We slide on the pre-prepared outer shell.

We pre-prepare approx. 80g of dry fofo_concrete mixture and the corresponding amount of water to it, which is approx. 17g. Right before casting we mix thoroughly.

We place the mold on the vibrating table, set a reasonable intensity of vibrations and gradually pour the mixture into the mold so that we do not clog the air exhaust channels. After filling we vibrate for at least another 1-2 minutes.

We leave the mold in a shady place at a temperature of **15-25°C**. For a stronger casting after 1-2 days of hardening we remove the outer shell and submerge the mold with the hardening mass into water for about 7-14 days.

Warnings and recommendations

Warnings

1. The specific used material must be carefully selected and compared with other options.
2. A mistake in calculations can manifest itself unfavorably and drastically.
3. The bond in the material degrades at high temperatures. This nozzle should be used single-use and for short-burn engines.

Recommendations